

MATH 1342 – Elementary Statistics – Week 5: Two-Variable Relationships and Linear Regression

Sep 21 – 23 | 2.4 Two Variable Relationships • 2.5 Two Quantitative Variables: Linear Regression | Fall 2026 • Section 019 • MW 7:30–8:50 PM • RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 5. Analyze relationships between two quantitative variables using correlation and linear regression.	1. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CO1, CO2) 2. Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r . (CO5) 3. Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CO5) 4. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CO5) 5. Use a regression equation to predict a response value, and compute and interpret a residual. (CO5) 6. Use technology to find the least-squares regression line and explain what “least squares” means. (CO5) 7. Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CO5)	e-Text: 2.4 Two Variable Relationships	MO1, MO2, MO3	Participation: Box Plots and Outliers – Mon: 1.5×IQR rule review; side-by-side boxplots (tea vs. coffee)	MO1	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 27, 11:59 PM)	MO1, MO2, MO3
		Slides: 2.4 Two Variable Relationships	MO1, MO2, MO3	Participation: Two Variable Relationships – Wed: NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier; Your Turn: spotting residuals	MO2, MO3, MO5	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM)	MO4–MO7
		e-Text: 2.5 Two Quantitative Variables: Linear Regression	MO4–MO7	Examples: baseball game length and commute-time regression lines (predict, interpret slope, find a residual)	MO4, MO5, MO6	Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice)	None
		Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1)	MO4, MO5				
		Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2)	MO6, MO7				
		Slides: 2.5 Two Quantitative Variables: Linear Regression	MO4–MO7				